

</> Asadullah Shafique

Agentic AI Engineer | Multi-Agent Systems Architect | Enterprise AI Strategist

Bridging AI potential with financial services reality through orchestration, compliance, and scalable adoption.

📍 Karachi, Pakistan 📞 +92-321-3771445 ✉️ asadullahshafique@hotmail.com

🔗 github.com/asadullah48 🌐 LinkedIn 🌐 Portfolio Medium

📁 Professional Summary

Agentic AI Engineer building multi-agent systems for regulated industries, backed by **25+ years** of operating a manufacturing business. Most agentic pilots in financial services die at the compliance review — not because the models are weak, but because a system that cannot show *why* it reached a conclusion cannot be signed off by a second line of defence. My work makes that reviewable: typed orchestration with an auditable route, a written constitution screened deterministically before any model call, and evaluation that reads the execution trace rather than trusting the prose.

Founder of Texcot Embroidery Sourcing House and creator of the **Textile ERP Platform**, targeting fabric mills, CMT units and garment exporters across Faisalabad, Sialkot, Gujranwala, Karachi and Lahore — so the domain intelligence is built from inside an industry rather than modelled from outside one. Completed **6 consecutive Panaversity Hackathons** (Bronze → Silver → Silver → Gold → Platinum → Agent Factory) with **85% code reusability** and **zero failed attempts**.

</> Agentic AI Mastery

Twelve competencies, four categories. Each cites the path that demonstrates it.

AGENTIC ORCHESTRATION

تنسيق الوكلاء

- › Engineering Multi-Agent Ecosystem Design (MAS)
backend/orchestration/
- › Designing Graph-Based Agent Workflows
LangGraph · Plan-Act-Verify
- › Implementing Agentic Coding & SDLC Standardization
CLAUDE.md · spec-first

CONNECTIVITY & PROTOCOLS

الاتصال والبروتوكولات

- › Engineering MCP Interoperability Layers
/mcp/server · FastMCP
- › Standardizing Read-Only Agent Tool Contracts
6 tools, read-only
- › Architecting Provider-Agnostic Fallback Ladders
SDK → LangGraph → static

ENTERPRISE & SCALE

المؤسسات والتوسع

- › Engineering Domain-Specific Intelligence (Finance)
FinAgent-Nexus · 94 tests
- › Architecting Domain-Specific ERP Intelligence (Textile)
CMT · live
- › Implementing Cloud-Native Scale-Out
k8s/ · 11 manifests

RELIABILITY & SAFETY

الموثوقية والأمان

- › Implementing Deterministic Guardrails
constitution/ · 4-of-4 blocked
- › Designing Feedback-Driven Evaluation Loops
evals/ · 3 suites
- › Engineering Context Integrity & Trace Auditability
orchestration/context.py

</> Technical Skills

LANGUAGES

TypeScript

JavaScript

Python

HTML/CSS

BACKEND

FastAPI

SQLAlchemy ORM

PostgreSQL

REST APIs

CLOUD & DEVOPS

Kubernetes

Docker

Dapr

Kafka

Helm

GitHub Actions

Vercel

Koyeb

Minikube

PLATFORMS & TOOLS

Git

Windows 11

WSL2

FRONTEND

Next.js 15

React 19

Tailwind CSS v4

shadcn/ui

AI & AGENTS

OpenAI Agents SDK

MCP Servers

Constitutional AI

RAG

OpenAI API

OBSERVABILITY

Prometheus

Grafana

Jaeger

Loki

Hackathon Achievements

Panaversity Hackathon Series: 6 Consecutive Wins · 85% Code Reuse · Zero Failed Attempts



Platinum 2025

Agent Factory (H5)

General Agent (Claude Code) manufactures Custom Agents (OpenAI Agents SDK). SKILL.md monetization. Digital FTE deployed on Kubernetes + Dapr. 117 - slide deck.

Claude Code

OpenAI SDK

MCP

SKILL.md

Dapr

K8s



Platinum 2025

Cloud-Native Deployment (H4)

Full K8s stack: Kubernetes + Docker + Dapr + Kafka + Prometheus/Grafana/Jaeger + Helm + GitHub Actions CI/CD.

K8s

Docker

Dapr

Kafka

Helm

Prometheus



Gold 2025

Advanced Todo (H3)

149 tests passing. Recurring todos, team collaboration, AI suggestions, triple-layer Constitutional AI (7 BLOCK + 5 FLAG patterns).

Next.js

FastAPI

PostgreSQL



Silver 2025

AI-Powered Todo (H2)

89 tests passing. Full-stack Constitutional AI todo. 70% code reuse from H1.

Next.js

TypeScript

FastAPI



Silver 2024

Course Companion FTE (H1)

Zero-Backend-LLM architecture. Digital FTE model. 70% code reuse from H0.

Next.js

TypeScript

OpenAI



Bronze 2024

Personal AI CTO (H0)

Flagship Projects

The Autonomous Agent Ecosystem [↗](#)

Shipped 

21 production platforms, one per problem domain

21 production multi-agent platforms, each with its own FastAPI gateway, test suite and public repository. 263 automated tests green in aggregate across the 21 repositories, all open source.

Agents SDK

FastAPI

Kubernetes

FinAgent - Nexus [↗](#)

Shipped 

Multi-Agent Financial Intelligence

Settles 8 of 14 compliance principles by arithmetic — a full screen in under 2ms with zero model calls, and 94 tests that pass with no API key and no network.

LangGraph

Constitutional AI

Python 3.12

AI TradeFlow [↗](#)

Shipped 

Inventory & Accounting for Wholesalers

90 automated tests including full-trade-cycle integration and agent golden-question suites with tool-citation assertions. Munshi AI ships 5 read-only tools behind a deterministic constitutional screen.

OpenAI Agents SDK

FastAPI

PostgreSQL

DevUnity Platform [↗](#)

Shipped 

This portfolio — and its own proof

A triage orchestrator over four specialists, a real MCP server on the official SDK, and a written constitution verified blocking 4 of 4 violations with no model reachable.

Agents SDK

FastMCP

Next.js 15

Bazaar [↗](#)

Shipped 

Unified B2B + B2C Marketplace

500+ verified sellers and 10K+ products across one codebase serving both retail checkout and a wholesale RFQ engine. PKR-native, three languages.

Next.js 15

FastAPI

Supabase

Textile ERP Platform [↗](#)

In build

Multi-tenant SaaS for Pakistan's textile heartland

Targets Faisalabad, Sialkot, Gujranwala, Karachi and Lahore. Order lifecycle, four auto-billing types, party ledgers and BOM inventory. In build.

Kubernetes

FastAPI

PostgreSQL

Agent Factory (H5) [↗](#)

Shipped 

A General Agent that manufactures Custom Agents

Two-tier architecture where Claude Code manufactures OpenAI Agents SDK agents; SKILL.md as portable intelligence units. Platinum tier, deployed on Kubernetes + Dapr.

Claude Code

SKILL.md

Dapr

</> Key Projects

Textile ERP Platform (Pakistan)

In Development

2024 – Present

Founder

Full-scale ERP targeting Pakistan's textile and garment industry, from Fabric Mills to CMT units to garment exporters.

- › Fabric Mill Module: Roll/lot management, weaving & knitting stage tracking, yarn inventory, imported fabric handling
- › CMT Core: Order lifecycle management, auto-billing (4 bill types), inventory with BOM, production sessions, dispatch tracking
- › Financial: Party ledgers, cash flow tracking, financial accounts for multi-city hubs (Faisalabad, Sialkot, Gujranwala, Karachi, Lahore)

Next.js 15

FastAPI

PostgreSQL

Vercel

Koyeb

🔗 Live Demo

Agent Factory (H5)

Panaversity Hackathon 2025

2025

Lead Developer

Two-tier agentic architecture: General Agent manufactures Custom Agents at scale.

- › General Agent (Claude Code) manufactures Custom Agent (OpenAI Agents SDK)
- › SKILL.md files as portable, reusable, monetizable intelligence units
- › Digital FTE pricing model targeting OpenAI Apps ecosystem (800M users)
- › Deployed on Kubernetes + Dapr

Claude Code

OpenAI SDK

MCP

SKILL.md

Dapr

Kubernetes

Professional Experience

Founder & CEO, CMT Stitching Operations

Texcot Embroidery Sourcing House

2020 – Present
Karachi, Pakistan

- › Engineered a multi-agent orchestration layer for financial-services research and Shari'ah screening, with separation of powers enforced structurally — no agent holds two of market data, weighting and verdict — rather than by prompt
- › Implemented MCP interoperability so agent capability is exposed as read-only tools on a protocol server, verified against a live client, instead of as bespoke HTTP endpoints
- › Architected deterministic guardrails from a written constitution, screened before any model call — verified blocking 4 of 4 violations with no model reachable, so enforcement survives a provider outage
- › Architected the Textile ERP Platform — order lifecycle, four auto-billing types, party ledgers and BOM inventory — to digitize Pakistan's textile value chain
- › Founded and operate a CMT stitching unit of 30–35 staff, running the full manufacturing lifecycle from sample fabric and design analysis through machine allocation, bulk production and inspection
- › Directed digital marketing across social channels, property portals and lead-generation funnels for textile and GCC real-estate clients

Marketing Manager

JK Embroidery

2016 – 2020
Karachi, Pakistan

- › Directed marketing and production planning for embroidery manufacturing operations
- › Engineered staff and equipment scheduling to clear production bottlenecks, implementing process improvements that held under peak order flow
- › Led a team of 30–35 staff, setting priorities against available resources and work-order flow

Country Manager / Buying Agent

Steven Berry

2003 – 2005
International

- › Managed and organized international shipments, providing clients guidance throughout procurement processes
- › Ensured products met established international quality standards across garment and textile categories
- › Conducted pre-production, in-production, and final inspection of apparel, accessories, and textile products

Digital Marketing Services

Full-spectrum digital marketing strategy for Dubai real estate, UAE construction, and Pakistani SMEs:

- › Social media management (Facebook, Instagram, X/Twitter, LinkedIn) + paid advertising campaigns
- › Property portal listings, lead generation funnels, and monthly performance reporting
- › Tailored strategies for GCC/expat/Pakistani diaspora investors; AED-priced service packages

Education & Certifications

Governor Sindh Initiative for Artificial Intelligence & Computing (Panaversity) Feb 2024 – Present
Advanced study in Agentic AI and Generative AI

Forward Deployed Engineer Training — Track B (Accelerated), 13 weeks:
governed Vertical Systems of Record, stateless MCP (rev. 2026-07-28), multi-agent research systems, and structured extraction pipelines with validation and human review.

TypeScript / JavaScript

React / Next.js

Python

Agent SDKs & AI frameworks

Loop / Harness / Graph architectures

EXPECTED

Sindh Board / University of Karachi — Programme certification

Panaversity — PCAR-F — Architect Foundations, aligned to Anthropic's CCAR-F blueprint

Anthropic — Claude Certified Associate – Foundations → Architect – Foundations

Alim (5-Year Islamic Studies Course)

Burooj Institute, Karachi

Associate Degree in Textile Technology

1997

Textile Institute of Pakistan (APTMA)

H.S.C Pre-Engineering

1994

Pakistan Shipowner's Government College, Karachi

S.S.C Science

1992

Ladybird Grammar School, Karachi